Edi Zhang

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Website | Linkedin | Github

Education

University of California, Los Angeles (UCLA) | Los Angeles, CA *Expected Graduation 2026*
BS, Applied Mathematics, Minor in Data Science Engineering, Specialization in Computing

- **GPA:** 3.82/4.0 — Dean's Honors List; ACT: 35
- **Core Math/EECS Coursework:** Real Analysis I/II, Dynamical Systems, Discrete Math, Differential Equations, Data Structures/Algorithms, Classical/Quantum Mechanics, Systems and Signals, Mathematical Modeling;
- **Machine Learning Coursework:** Neural Networks/Deep Learning I/II (Graduate), Applied Numerical Computing, Abstract Linear Algebra (Honors), Computational Imaging (Graduate), Probability, Statistics;
- **Skills:** Python, C++, MATLAB, Data Analysis (NumPy, Pandas, Matplotlib), Machine Learning (PyTorch, Tensorflow, Sklearn), SQL, Julia, ReactJS (HTML/CSS/JS), Excel, Indesign

Work Experience

Capital One | Incoming Software Engineer Intern | Richmond, VA *Jun 2025 - Aug 2025*

Cleveland Clinic Lerner Research Institute | Researcher | Cleveland, OH *Jun 2024 - Mar 2025*

- Researched in the Advanced Musculoskeletal Imaging Lab on mathematical modeling of tibiofemoral joint using Generalized Procrustes and Principal Component statistical analyses techniques to analyze directions of bone shape variance
- Constructed topological surfaces by processing 275 MRI masks using Marching Cubes and Laplacian Smoothing
- Implemented variance decomposition on five surface models to obtain hyper-dimensional eigenvectors representing principal directions of shape variance to ultimately improve characterization of knee osteoarthritis development
- Evaluated model by reconstruction accuracy, compactness, osteoarthritis severity stratification, and variance directions
- Presented abstract at 2025 Orthopaedic Research Society Annual Conference and preparing manuscript for publishing

UCLA Department of Mathematics | Course Reader | Los Angeles, CA *Sep 2024 - Present*

- Generating XML files and Python scripts to efficiently score homework assignments for C++ programming course
- Grading assignments for over 90 students, ensuring timely feedback and detailed evaluations for coding projects
- Assisting TAs/LAs with clarifying programming concepts and offering support on debugging and optimizing code

Extracurricular Activities

The Bruin Group Consulting | Senior Consultant | Los Angeles, CA *Jan 2024 - Present*

- Engaging in pro-bono strategy/marketing projects through premier consultancy (4.5% acceptance rate) for startups
- Spearheaded market research on M&A strategies for a multi-billion dollar top-5 U.S. aerospace and defense company
- Collaborated with vegan snack startup to research market entry analysis and pricing strategies towards college students

Bruin Sport Analytics | Data Journalist | Los Angeles, CA *Jan 2024 - Jun 2024*

- Leveraged in-depth data analysis and statistical techniques to publish data-driven articles on sports performances
- Presented and pitched article ideas and data processing/analysis methods at workshops for data science fundamentals
- Underwent 10-week statistics/data science course with NumPy, Pandas, Tensorflow, Matplotlib, Sklearn, PyTorch

Undergraduate Mathematics Student Association | Active | Los Angeles, CA *Sep 2023 - Present*

- Volunteering in student tutoring and mentorship groups, Integration Bee, and organizing professor research talks
- Recruiting new members and promoting club events, socials, competitions, and presentations for students on campus

Projects

EMG Classification | *Python, PyTorch* | Github *May 2024 - Jun 2024*

- Tested EMG Encoder to measure keystroke classification accuracy using various architectures including CNNs (75.41%), LSTMs (77.52%), GRUs (77.05%), CNN + LSTM (79.47%), and CNN + GRU (80.16%) on emg2qwerty datatset
- Achieved 80.55% accuracy by fine-tuning using CNN + GRU + Gaussian Noise and analyzed other combinations of fine-tuning methods including Frequency masks, Downsampling, and Random Cropping to optimize performance
- Analyzed test and validation accuracy and model compactness across architectures to optimize generalizability

Temple of Doom | *C++* | Github *May 2024 - Jun 2024*

- Multiple-level single player game managing 50+ actors and objects each player move using dynamic memory allocation
- Uses polymorphism and inheritance to create 15 types of actors/objects which manage game interactions each move
- Randomly generates maps, actors, and objects at runtime and uses recursion to determine optimal moves for monsters

Freestyle Race Strategies | *Python* | Article *Jan 2024 - Feb 2024*

- Analyzed race splits and time progressions across freestyle races to predict potential times among top swimmers
- Web-scraped data from 1000+ swimmers to construct linear regression modeling front and back half race splits

Additional Information

Competitions: BCG Case Competition Finalist, ACM Hackathon, ACM ICPC, ASA Datafest, Integration Bee, ICCA
Achievements: AIME Qualifier (2-time), AP Scholar with Distinction, National Merit Commended Scholar, Monte Vista Athletic Booster Scholarship, College Club Swimming Nationals Qualifier, USA Swimming CA/NV Sectionals Finalist
Volunteering: Campus Barber, Awaken A Cappella, Division I Swim Team Volunteer, UCLA Volunteer Day
Interests: Blitz Chess (2200), Swimming, Floral Designing, Barbering, Musical Theater, Speedcubing, Jazz Piano,